

DSA210 Final Project Report

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Identifying the Optimal Study Time for Maxi influenced by many different factors such as study habits, preparation strategies, family background, and socioeconomic conditions. Among these variables, study time is one of the most commonly discussed indicators of student success. However, studying for longer hours does not always guarantee better results, which raises an important question:

Is there an optimal amount of study time associated with higher academic performance?

The purpose of this project is to investigate how study-related factors affect students' exam scores using statistical analysis and machine learning methods. In particular, the project focuses on weekly study hours and test preparation while also examining the effects of several demographic and behavioral variables.

The project combines exploratory data analysis (EDA), hypothesis testing, and predictive modeling to better understand both the statistical and predictive relationships within the dataset.

Research Question

How do study-related variables such as weekly study hours and test preparation affect students' academic performance, and can these variables be used to accurately predict exam scores?

To answer this question, the project explores:

- whether study hours are significantly related to exam performance,
- whether test preparation improves scores,
- which variables contribute most strongly to prediction,
- and whether machine learning models can successfully predict student performance.

Dataset Description

The project uses the “Student Test Scores: Extended Dataset” obtained from Kaggle. The dataset contains more than 30,000 student observations and includes both demographic and educational variables.

Some of the main variables in the dataset are:

- Gender
- Ethnic Group
- Parent Education Level
- Lunch Type
- Test Preparation Course
- Parent Marital Status
- Weekly Study Hours
- Number of Siblings
- Transportation Method

- Sports Participation
- Math, Reading, and Writing Scores

To represent overall academic performance, a new variable called **ExamScore** was created by averaging the math, reading, and writing scores.

This variable was used as the target variable in the machine learning models.

Data Preprocessing

Before starting the analysis, several preprocessing steps were applied.

First, missing values in categorical variables were handled using mode imputation. Since most missing variables were categorical, replacing them with the most frequent category helped preserve the size of the dataset without introducing major changes.

In addition, a new feature called **ExamScore** was created by averaging the three exam scores. This made it easier to evaluate overall academic performance with a single target variable.

After preprocessing, the dataset was ready for exploratory analysis and machine learning applications.

Exploratory Data Analysis (EDA)

EDA was conducted to better understand the relationships between variables before applying statistical tests and machine learning models.

Weekly Study Hours

The distribution of weekly study hours showed that most students were concentrated in moderate study groups, while relatively fewer students studied for very high numbers of hours.

Grouped averages also revealed a clear trend: students who studied more generally achieved higher exam scores.

However, the increase in scores did not appear perfectly linear. The improvement became smaller at higher study levels, suggesting that the relationship between study time and performance may experience diminishing returns.

This observation supports the idea that there may be an “optimal” study range rather than unlimited benefits from continuously increasing study time.

Test Preparation

Students who completed the test preparation course consistently achieved higher average exam scores than students who did not.

The difference between the two groups was visually noticeable and later confirmed with hypothesis testing.

Other Variables

Variables such as parental education, lunch type, and marital status also showed some relationship with exam scores. However, these effects appeared weaker than study-related variables.

Overall, the EDA results suggested that effort-related factors are more strongly associated with academic performance than demographic variables.

Hypothesis Testing

To statistically evaluate the patterns observed during EDA, hypothesis tests were applied.

Effect of Test Preparation

An independent two-sample t-test was conducted to compare the average exam scores of students who completed the test preparation course and those who did not.

Hypotheses

- **H0:** There is no difference in average exam scores between the two groups.
- **H1:** There is a significant difference in average exam scores between the two groups.

Results

- T-statistic ≈ 39.10
- P-value ≈ 0.0

Interpretation

Since the p-value is far below the significance level, the null hypothesis was rejected. The results indicate that students who completed the preparation course achieved significantly higher exam scores.

This suggests that structured preparation has a positive relationship with academic performance.

Effect of Weekly Study Hours

A one-way ANOVA test was used to compare exam scores across different study-hour groups.

Hypotheses

- **H0:** All study-hour groups have the same average exam score.
- **H1:** At least one group has a different average exam score.

Results

- F-statistic ≈ 75.65
- P-value $\approx 1.67e-33$

Interpretation

The extremely small p-value indicates that there are statistically significant differences between study-hour groups.

Students who studied more generally performed better on exams.

However, although study hours are statistically significant, this does not automatically mean that they are strong predictors of academic performance. This distinction becomes clearer in the machine learning section.

Machine Learning Models

Machine learning models were applied to predict students' overall exam performance using the available features.

The following models were tested:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

The models were evaluated using:

- RMSE (Root Mean Squared Error)
- R^2 Score

- Cross-validation
- Residual analysis

Model Comparison

Linear Regression

Linear Regression produced the best overall performance among the tested models.

- RMSE $\approx$ 12.36
- $R^2 \approx$ 0.26

Although the model performed better than the others, the R^2 value remained relatively low. This means that the variables in the dataset explain only a limited portion of the variation in exam scores.

Decision Tree Regressor

The Decision Tree model performed poorly and produced a negative R^2 value.

This indicates that the model generalized badly and performed worse than simply predicting the average score.

Random Forest Regressor

The Random Forest model slightly improved over the Decision Tree but still showed limited predictive power.

Its R^2 score remained relatively low, suggesting that model complexity alone was not enough to substantially improve prediction.

Cross-Validation and Hyperparameter Tuning

Cross-validation produced relatively consistent R^2 scores around 0.25 across folds.

This suggests that the models were reasonably stable and did not suffer from severe overfitting.

Hyperparameter tuning was also applied to optimize the Decision Tree model. The best-performing tree depth was found to be around 5.

However, tuning did not significantly improve model performance.

This indicates that the main limitation comes from the dataset itself rather than the model configuration.

Feature Importance Analysis

Feature importance analysis produced several interesting findings.

The number of siblings appeared as the most influential variable in the model.

In addition, lunch type, parental marital status, and transportation method showed moderate predictive importance.

Study-related variables such as weekly study hours also contributed to prediction, but they were not dominant features.

Interestingly, test preparation did not rank among the strongest predictive variables even though it was highly significant in earlier statistical tests.

This highlights an important distinction between:

statistical significance and predictive importance.

A variable may show a meaningful statistical relationship while still contributing limited predictive value in machine learning models.

The unusually high importance of the number of siblings may reflect indirect socioeconomic relationships or possible noise within the dataset rather than a direct causal effect.

Residual Analysis

Residual plots showed that prediction errors were generally scattered around zero without a strong systematic pattern.

This suggests that the models were not heavily biased.

However, the residual spread remained relatively wide, indicating that the models struggled to accurately predict individual exam scores.

In other words, the models were able to capture general trends but could not reliably predict exact student outcomes.

Overall Interpretation

The combined results from EDA, hypothesis testing, and machine learning provide several important insights.

First, study-related variables such as weekly study hours and test preparation clearly show statistically significant relationships with academic performance.

Students who study more and complete preparation courses generally achieve higher exam scores.

However, the machine learning results demonstrate that these variables alone are not sufficient to build highly accurate predictive models.

Even the best-performing model explained only a limited portion of score variation.

This suggests that academic performance depends on many additional factors that are not included in the dataset, such as:

- motivation,
- learning quality,
- sleep habits,
- attendance,
- and personal ability.

One of the most important findings of this project is therefore the distinction between: finding statistically significant relationships and building strong predictive models.

Limitations

This project has several limitations.

The dataset is observational rather than experimental, meaning that causal conclusions cannot be made.

In addition, some variables such as study hours may contain self-report bias.

The dataset also lacks several potentially important variables related to psychology, learning behavior, and personal motivation.

These missing variables likely contribute to the relatively low predictive performance of the machine learning models.

Future Work

Future studies could improve this project by:

- including behavioral and psychological variables,
- using larger or more diverse datasets,
- applying additional machine learning techniques,
- and investigating nonlinear relationships between study habits and academic performance.

It may also be useful to examine study efficiency and study quality rather than only study duration.

Conclusion

This project investigated how study-related factors affect academic performance using both statistical analysis and machine learning methods.

The results consistently showed that students who study more and complete test preparation courses tend to perform better academically.

At the same time, the machine learning models demonstrated that these variables alone are not enough to accurately predict exam performance.

Overall, the findings suggest that academic success is influenced by a combination of effort, background, environment, and individual factors rather than a single dominant variable.

The project therefore provides both statistical evidence for the importance of study habits and practical insight into the limitations of predictive modeling in educational datasets.